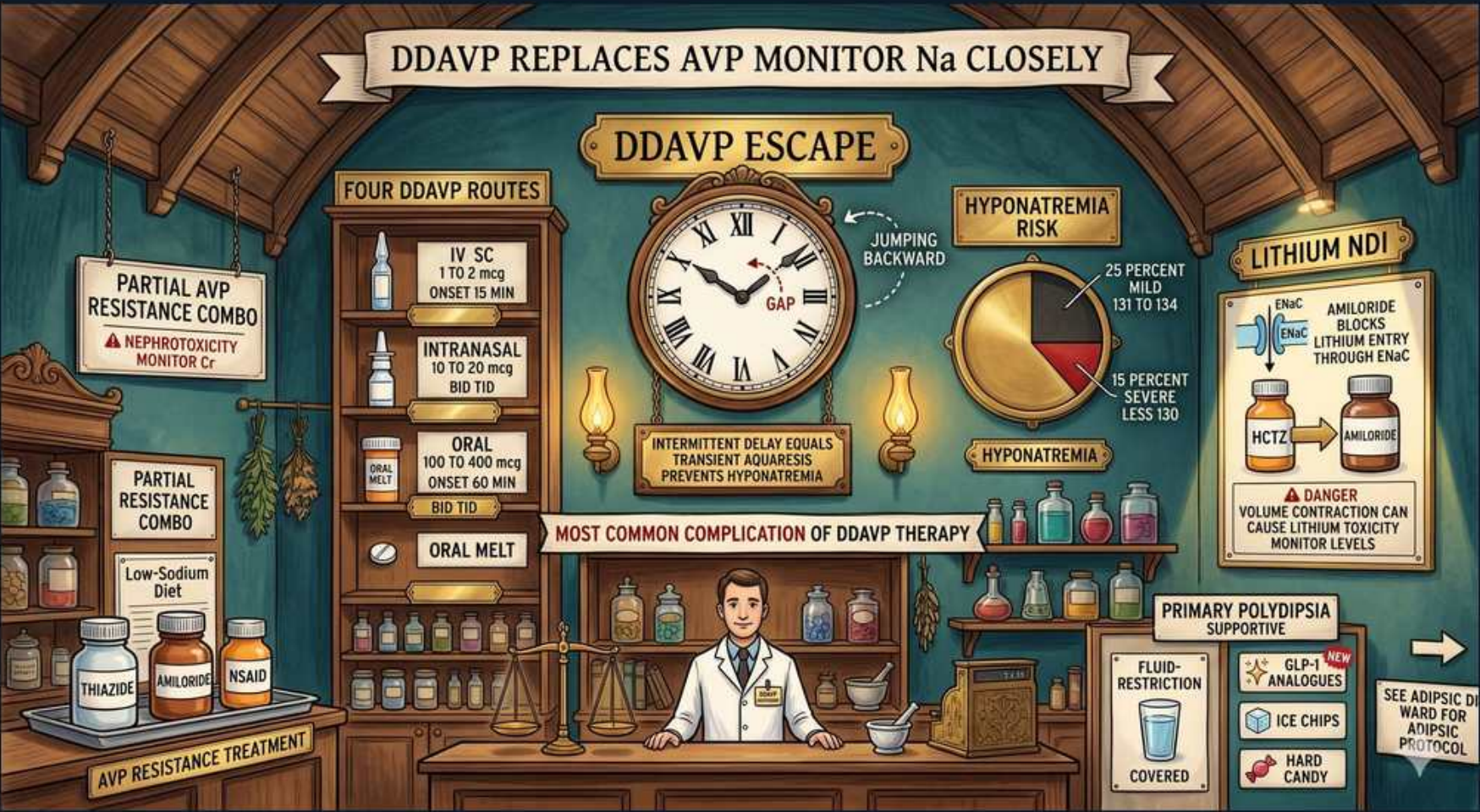


Pituitary/DI/SIADH — Diabetes insipidus treatment



1. DDAVP replaces AVP

WHAT YOU SEE

Banner: DDAVP replaces AVP, monitor Na closely

WHAT IT ENCODES

Central DI (AVP deficiency) treatment is desmopressin (DDAVP), a V2-selective AVP analog that replaces the missing hormone and restores urine concentration. Because DDAVP forces water retention, the chief hazard is dilutional HYPONATREMIA, so serum sodium must be monitored and a daily 'aquaresis window' allowed. Ensure free access to water and treat any coexisting anterior-pituitary deficits (especially cortisol, which can mask DI until replaced).

BOARD TRAP

WRONG: central DI is treated with thiazide diuretics. Discriminator — thiazides treat NEPHROGENIC DI; central DI is a hormone-deficiency state corrected by replacing the hormone (DDAVP). Giving only a thiazide in central DI leaves the patient undertreated.

2. Four DDAVP routes

WHAT YOU SEE

An apothecary cabinet headed FOUR DDAVP ROUTES, its shelves stocked with IV/SC vial, intranasal spray, oral tablet, and oral-melt — four ways to deliver the same replacement hormone.

WHAT IT ENCODES

DDAVP is available in four routes — IV/SC, intranasal spray, oral tablet, and oral-disintegrating (sublingual) melt — letting therapy be tailored to acuity and setting. Onset, potency, and bioavailability differ markedly: parenteral is most potent/fastest; oral has the lowest bioavailability. Doses are NOT interchangeable milligram-for-milligram across routes, so each route is titrated independently.

BOARD TRAP

WRONG: DDAVP is only available as an IV formulation. Discriminator — intranasal and oral forms dominate outpatient maintenance; IV/SC is reserved for acute/inpatient or perioperative use. Assuming IV-only would miss the practical maintenance options patients actually use.

3. IV/SC route

WHAT YOU SEE

IV SC 1-2 mcg onset 15 min

WHAT IT ENCODES

IV or subcutaneous DDAVP 1–2 mcg has the FASTEST onset (~15 min) and highest potency, making it the choice for acute management, perioperative central DI, or when enteral/intranasal routes are unreliable. Because potency is high, sodium and urine output should be watched closely to avoid over-correction and hyponatremia.

BOARD TRAP

WRONG: oral DDAVP works faster than the IV/SC route. Discriminator — IV/SC is the fastest-onset, highest-potency route (~15 min); oral has the slowest onset and lowest bioavailability. For an acutely polyuric inpatient, choose parenteral.

4. Intranasal route

WHAT YOU SEE

Intranasal 10-20 mcg BID/TID

WHAT IT ENCODES

Intranasal DDAVP 10–20 mcg dosed BID–TID is a common outpatient maintenance route with intermediate onset and bioavailability. Absorption can be erratic with rhinitis, nasal congestion, or upper-respiratory infection, which may require dose adjustment or temporary switch to oral/parenteral.

BOARD TRAP

WRONG: intranasal DDAVP absorption is unaffected by nasal pathology. Discriminator — mucosal inflammation (URI, allergic rhinitis) reduces and makes intranasal absorption unreliable; a patient whose DI 'breaks through' during a cold may simply need a route change, not a higher chronic dose.

5. Oral / oral melt

WHAT YOU SEE

Oral 100–400 mcg, oral melt

WHAT IT ENCODES

Oral DDAVP (100–400 mcg) and the sublingual orally-disintegrating melt are convenient but have the LOWEST bioavailability (oral tablet only ~1–5%), so doses are much larger than intranasal/parenteral and are titrated to symptom control and serum Na. Food reduces tablet absorption, so consistent timing relative to meals matters.

BOARD TRAP

WRONG: oral DDAVP can use the same microgram dose as the intranasal or IV form. Discriminator — because oral bioavailability is very low, oral doses are roughly 10–20× the intranasal dose; treating route doses as interchangeable would massively underdose an oral patient.

6. Intermittent delay = aquaresis

WHAT YOU SEE

Clock with gap, intermittent delay equals transient aquaresis

WHAT IT ENCODES

A key safety strategy is to deliberately allow a daily 'escape' or aquaresis window — let one DDAVP dose wear off before the next so the kidney can excrete any retained free water. This brief return of dilute urine each day prevents progressive water accumulation and dilutional hyponatremia. Patients are counseled to drink to thirst, not on a fixed schedule, while on DDAVP.

BOARD TRAP

WRONG: DDAVP should be dosed to keep the urine concentrated 24 hours a day. Discriminator — continuous around-the-clock antidiuresis without an escape window is the classic setup for iatrogenic hyponatremia; intentional intermittent dosing (planned aquaresis) is protective.

7. DDAVP escape / hyponatremia

WHAT YOU SEE

DDAVP escape title + hyponatremia risk pie chart

WHAT IT ENCODES

Hyponatremia is the MOST COMMON complication of DDAVP therapy, driven by water retention especially if the patient keeps drinking beyond thirst. Risk is highest in young children, the elderly, and anyone given supplemental IV fluids. Mitigation: lowest effective dose, daily aquaresis window, drink-to-thirst counseling, and periodic serum Na checks.

BOARD TRAP

WRONG: hyperNATREMIA is the main complication of DDAVP therapy. Discriminator — untreated central DI causes hypernatremia, but the TREATMENT (DDAVP) causes water retention and therefore hypoNATREMIA; falling Na on DDAVP means hold the dose, not give more.

8. Hyponatremia severity

WHAT YOU SEE

Pie: 25% mild 131-134, 15% severe <130

WHAT IT ENCODES

DDAVP-associated hyponatremia is usually MILD (Na ~131–134, roughly a quarter of treated patients) but can be SEVERE (<130, ~15%) and symptomatic. Management is to hold the next DDAVP dose, restrict free water, and recheck sodium; severe symptomatic cases follow the standard cautious hyponatremia-correction limits to avoid osmotic demyelination.

BOARD TRAP

WRONG: DDAVP-induced hyponatremia is always trivial and never needs intervention. Discriminator — a meaningful minority fall below 130 and can become symptomatic (headache, nausea, seizures); a falling Na warrants dose hold and water restriction, not reassurance alone.

9. Lithium NDI panel

WHAT YOU SEE

Lithium NDI box with amiloride + HCTZ

WHAT IT ENCODES

Nephrogenic DI (AVP resistance) treatment: remove/treat the cause (stop or reduce lithium if possible, correct hypercalcemia and hypokalemia), then reduce solute/water load with a LOW-SALT, low-protein diet plus a thiazide (HCTZ), amiloride, and/or an NSAID. DDAVP does NOT work here because the defect is renal resistance, not hormone deficiency.

BOARD TRAP

WRONG: nephrogenic DI is treated with desmopressin. Discriminator — the kidney cannot respond to AVP in nephrogenic DI, so DDAVP fails; treatment instead lowers urine output via thiazide + low-salt diet ± amiloride/NSAID and by correcting the underlying drug/metabolic cause.

10. Amiloride for lithium NDI

WHAT YOU SEE

Amiloride blocks lithium entry through ENaC

WHAT IT ENCODES

Amiloride is FIRST-LINE for lithium-induced nephrogenic DI: it blocks the apical ENaC sodium channel in collecting-duct principal cells, the very channel through which lithium enters the cell to downregulate AQP2. By preventing lithium uptake, amiloride protects the cell and may even let the patient continue lithium when no alternative mood stabilizer exists.

BOARD TRAP

WRONG: amiloride worsens lithium toxicity. Discriminator — by blocking ENaC, amiloride REDUCES lithium entry into principal cells (protective) without causing the volume contraction that raises serum lithium; it is thiazides (volume contraction) that can raise lithium levels.

11. HCTZ / thiazide

WHAT YOU SEE

HCTZ in nephrogenic box; thiazide bottle bottom-left

WHAT IT ENCODES

Thiazides (e.g., HCTZ) PARADOXICALLY reduce urine output in DI: by blocking the distal Na-Cl cotransporter they cause mild volume contraction, which increases proximal sodium and water reabsorption and delivers less fluid distally. They work best paired with a low-sodium diet, which augments the volume-contraction effect.

BOARD TRAP

WRONG: thiazides increase urine output in DI. Discriminator — although diuretics normally raise urine output, in DI the thiazide-induced mild volume contraction enhances proximal reabsorption and DECREASES urine volume; this counterintuitive 'paradoxical antidiuresis' is a favorite exam point.

12. Volume contraction / Li toxicity

WHAT YOU SEE

Danger: volume contraction can raise lithium levels

WHAT IT ENCODES

Thiazide-induced volume contraction increases proximal tubular reabsorption of sodium AND lithium (lithium is handled like sodium), so it can RAISE serum lithium into the toxic range. When a thiazide is used in lithium nephrogenic DI, lithium levels and renal function must be monitored, which is part of why amiloride is preferred.

BOARD TRAP

WRONG: thiazides lower serum lithium levels. Discriminator — thiazides cause volume contraction that increases proximal lithium reabsorption, RAISING lithium and risking toxicity; amiloride (which blocks lithium uptake without volume contraction) is the safer first choice.

13. Partial resistance combo

WHAT YOU SEE

Partial AVP resistance combo: low-salt, thiazide, amiloride, NSAID

WHAT IT ENCODES

For partial nephrogenic DI, combine therapies for additive urine-reducing effect: low-sodium (and low-protein) diet to cut solute load, a thiazide for volume contraction, amiloride to block lithium/ENaC, and an NSAID (indomethacin) — prostaglandins normally antagonize AVP, so blocking them enhances water reabsorption. Watch NSAID renal and GI toxicity with chronic use.

BOARD TRAP

WRONG: NSAIDs have no role in nephrogenic DI because the kidney can't respond to AVP anyway. Discriminator — NSAIDs reduce renal prostaglandins (which oppose AVP and promote water excretion), so they reduce urine output independent of the AVP-resistance defect; they are a legitimate adjunct, dosed cautiously for nephrotoxicity.

14. Primary polydipsia supportive

WHAT YOU SEE

Primary polydipsia supportive: fluid restriction, ice chips, hard candy

WHAT IT ENCODES

Primary polydipsia is managed with fluid restriction and behavioral/symptomatic measures for dry mouth (ice chips, hard candy/sugarless gum) plus treating the underlying psychiatric disorder. Desmopressin is AVOIDED because the patient's AVP axis is intact and they continue drinking, so DDAVP would precipitate water intoxication and hyponatremia.

BOARD TRAP

WRONG: give desmopressin to a patient with primary polydipsia. Discriminator — these patients already over-retain water from drinking; adding DDAVP causes severe dilutional hyponatremia. Correct management is fluid restriction plus addressing the compulsive-drinking/psychiatric cause.